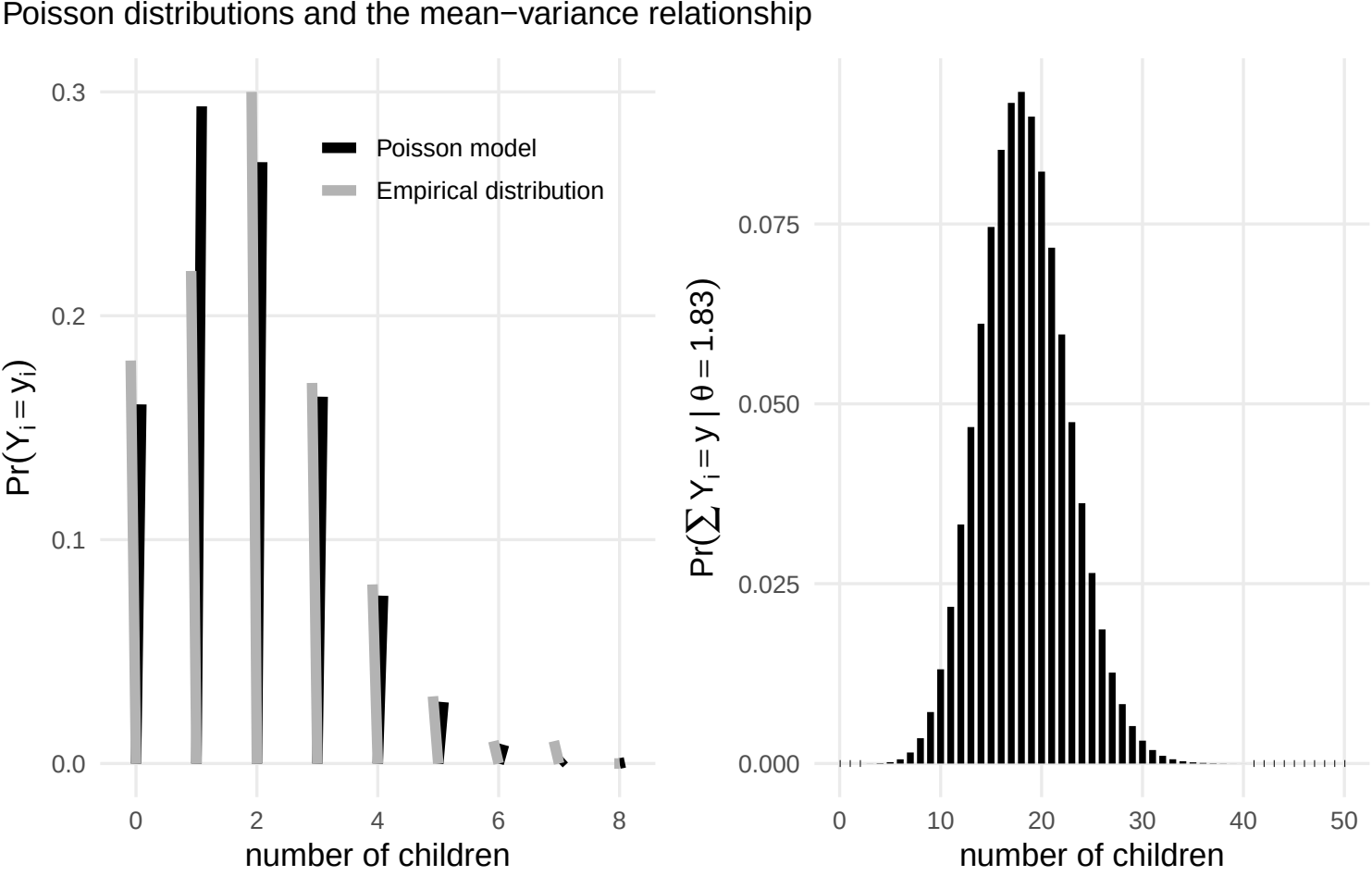

Left: Poisson pmf with mean theta = 1.83 (black), overlaid with the empirical distribution (grey).
Right: The sum of 10 independent Poisson(1.83) variables follows Poisson(18.3).
The increased spread illustrates the mean−variance relationship: larger means imply larger variances.